Summary Note: Electricity and Magnetism

1. Forces in Nature

There are four fundamental forces in nature:

- **Gravitational force** – attraction between masses
- **Electromagnetic force** – acts between charges
- **Strong nuclear force** – binds protons and neutrons
- **Weak nuclear force** – responsible for radioactive decay

2. Electrostatics

Electrostatics is the study of **charges at rest**.

Electric Charge

Types:

- Positive (+)
- Negative (−)

Properties:

- Like charges repel
- Unlike charges attract
- Charge is conserved

Methods of Charging

- Friction
- Conduction
- Induction

3. Coulomb's Law

Force between two charges:

$F = kq_1q_2 / r^2$

Where:

- F = force
- q_1, q_2 = charges
- r = distance between charges
- k = constant

Principle of Superposition

Total force = vector sum of all individual forces.

4. Electric Field

Region where a charge experiences force.
Formula:
E = F / q
Unit: N/C or V/m

5. Electric Potential

Work done per unit charge.
V = W / q
Unit: Volt (V)

6. Gauss's Law

Electric flux through a closed surface depends on enclosed charge.
Useful for symmetrical charge distributions.

7. Capacitance

Ability to store charge.
C = Q / V
Unit: Farad (F)
Capacitor stores electrical energy.

8. Electric Dipole

Two equal and opposite charges separated by distance.
Dipole moment:
p = qd

9. Energy in Electric Field

Energy stored in capacitor:
$U = \frac{1}{2}CV^2$

10. Conductors and Insulators

Conductors

Allow current flow
Examples: copper, aluminium

Insulators

Resist current flow
Examples: rubber, plastic

DC Circuits

11. Current

Rate of flow of charge.

$I = Q / t$

Unit: Ampere (A)

12. Voltage

Potential difference between two points.

Unit: Volt

13. Resistance

Opposition to current flow.

$R = V / I$

Unit: Ohm (Ω)

14. Ohm's Law

At constant temperature:

$V = IR$

15. Resistor Combinations

Series

$R = R_1 + R_2 + R_3$

Parallel

$1/R = 1/R_1 + 1/R_2 + 1/R_3$

16. DC Circuit Analysis

Use:

- Ohm's law
- Kirchhoff's laws
- Power law: $P = VI$

Magnetism

17. Magnetic Field

Region where magnetic force acts.

Unit: Tesla (T)

18. Lorentz Force

Force on moving charge in magnetic field:

$F = qvB \sin\theta$

19. Biot–Savart Law

Calculates magnetic field due to current element.

20. Ampere's Law

Relates magnetic field to current producing it.

21. Magnetic Dipole

A magnet behaves like dipole with north and south poles.

22. Dielectrics

Insulating materials placed between capacitor plates to increase capacitance.

23. Energy in Magnetic Field

Energy stored in inductor:

$U = \frac{1}{2}LI^2$

24. Electromotive Force (EMF)

Energy supplied per unit charge by source.

Unit: Volt

Examples:

- Battery
- Generator

Electromagnetic Induction

25. Electromagnetic Induction

Production of emf due to changing magnetic flux.

26. Self and Mutual Inductance

Self Inductance

Coil induces emf in itself.

Mutual Inductance

One coil induces emf in another.

27. Faraday's Law

Induced emf is proportional to rate of change of magnetic flux.

28. Lenz's Law

Induced current opposes the cause producing it.

29. Transformers

Step-Up Transformer

Increases voltage

Step-Down Transformer

Decreases voltage

30. Maxwell's Equations

Four equations describing electricity and magnetism.

They explain:

- Electric fields
- Magnetic fields
- Electromagnetic waves

31. Electromagnetic Oscillations and Waves

Oscillating electric and magnetic fields produce waves.

Examples:

- Radio waves
- Light
- X-rays

AC Circuits

32. Alternating Current (AC)

Current that changes direction periodically.
Common in household supply.

33. AC with Components

Resistor

Voltage and current in phase.

Inductor

Current lags voltage.

Capacitor

Current leads voltage.